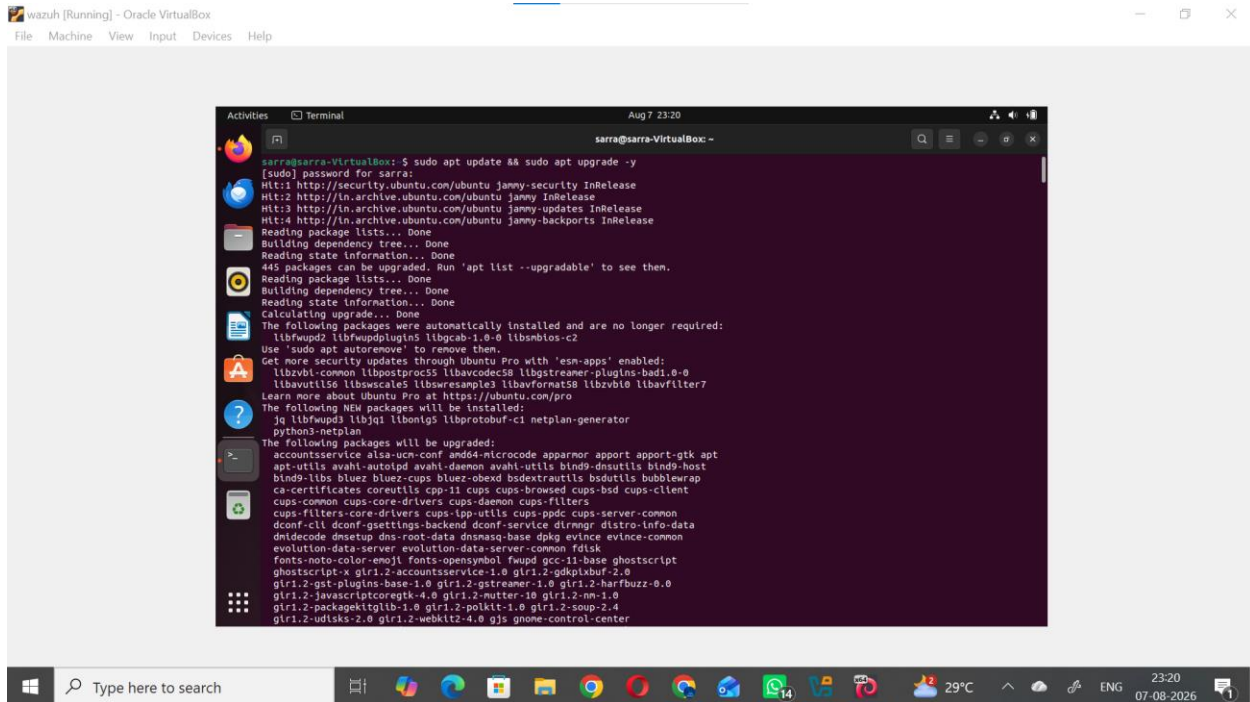


Wazuh SIEM with File Integrity Monitoring

Results

Name: Sarranya R

Wazuh manager on ubuntu



```
sarra@sarra-VirtualBox: ~$ sudo apt update && sudo apt upgrade -y
[sudo] password for sarra:
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
445 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libsnbios-c2
Use 'sudo apt autoremove' to remove them.
Get more security updates through Ubuntu Pro with 'esm-apps' enabled:
  libzvt1-common libpostproc5 libavcodec58 libgstreamer-plugins-bad1.0-0
  libavutil56 libswscale5 libswresample3 libavformat58 libzvt0 libavfilter7
Learn more about Ubuntu Pro at https://ubuntu.com/pro
The following NEW packages will be installed:
  jq libfwupd2 libjq libonig libprotobur-c1 netplan-generator
  python3-netplan
The following packages will be upgraded:
  accountsservice alsa-ucm-conf amd64-microcode apparmor apport apport-gtk apt
  apt-utils avahi-autoipd avahi-daemon avahi-utils bind9-dnsutils bind9-host
  bind9-libs bluez bluez-cups bluez-obexd bsdxtrautils bsdtails bubblewrap
  ca-certificates coreutils cpp-11 cups cups-browsed cups-bsd cups-client
  cups-common cups-core-drivers cups-daemon cups-filters
  cups-filters-core-drivers cups-lpp-utils cups-ppdc cups-server-common
  dconf-cli dconf-gsettings-backend dconf-service dirnagr distro-info-data
  dnidecode dnsetup dns-root-data dnsmasq-base dpkg evince evince-common
  evolution-data-server evolution-data-server-common fdisk
  fonts-noto-color-emoji fonts-opensymbol fwupd gcc-11-base ghostscript
  ghostscript-x glib1.2-accountsservice-1.0 glib1.2-gdkpixbuf-2.0
  glib1.2-gst-plugins-base-1.0 glib1.2-gstreamer-1.0 glib1.2-harfBuzz-0.0
  glib1.2-javascriptcoregtk-4.0 glib1.2-mutter-10 glib1.2-mn-1.0
  glib1.2-packagekit-glib-1.0 glib1.2-polkit-1.0 glib1.2-soup-2.4
  glib1.2-udisks-2.0 glib1.2-webkit2-4.0 gjs gnome-control-center
```

```
Activities Terminal Aug 7 23:21 sarra@sarra-VirtualBox: ~  
Running hooks in /etc/ca-certificates/update.d...  
done.  
Processing triggers for intransf-tools (0.14ubuntu13.5) ...  
update-intransf: Generating /boot/initrd.img-6.8.0-13-generic  
sarra@sarra-VirtualBox:~$ sudo apt install curl unzip gnupg apt-transport-https lsb-release software-properties-common -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
lsb-release is already the newest version (11.1.0ubuntu4).  
lsb-release set to manually installed.  
gnupg is already the newest version (2.2.27-3ubuntu2.5).  
gnupg set to manually installed.  
software-properties-common is already the newest version (0.99.22.9).  
software-properties-common set to manually installed.  
unzip is already the newest version (6.0-26ubuntu3.2).  
unzip set to manually installed.  
The following packages were automatically installed and are no longer required:  
libfwpd2 libfwpdplugins libgcb-1.0-0 libsnbios-c2  
Use 'sudo apt autoremove' to remove them.  
The following NEW packages will be installed:  
apt-transport-https curl  
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.  
Need to get 195 kB of archives.  
After this operation, 626 kB of additional disk space will be used.  
Get:1 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 apt-transport-https all 2.4.14 [1,510 B]  
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 curl amd64 7.81.0-1ubuntu1.25 [194 kB]  
Fetched 195 kB in 3s (75.1 kB/s)  
Selecting previously unselected package apt-transport-https.  
(Reading database ... 203201 files and directories currently installed.)  
Preparing to unpack ./apt-transport-https-2.4.14_all.deb ...  
Unpacking apt-transport-https (2.4.14) ...  
Selecting previously unselected package curl.  
Preparing to unpack ./curl-7.81.0-1ubuntu1.25 amd64.deb ...  
Unpacking curl (7.81.0-1ubuntu1.25) ...  
Setting up apt-transport-https (2.4.14) ...  
Setting up curl (7.81.0-1ubuntu1.25) ...  
Processing triggers for man-db (2.10.2-1) ...  
sarra@sarra-VirtualBox:~$ curl -LO https://packages.wazuh.com/4.14/wazuh-install.sh  
% Total % Received % Xferd Average Speed Time Time Current  
Dload Upload Total Spent Left Speed
```

```
Activities Terminal Aug 7 23:21 sarra@sarra-VirtualBox: ~  
100 203k 100 203k 0 0 618k 0 --:--:-- --:--:-- 620k  
sarra@sarra-VirtualBox:~$ ls -lh wazuh-install.sh  
-rw-rw-r-- 1 sarra sarra 203k Aug 7 23:02 wazuh-install.sh  
sarra@sarra-VirtualBox:~$ chmod +x wazuh-install.sh  
sarra@sarra-VirtualBox:~$ sudo ./wazuh-install.sh -a  
07/08/2026 23:03:46 INFO: Starting Wazuh installation assistant. Wazuh version: 4.14.7  
07/08/2026 23:03:46 INFO: Verbose logging redirected to /var/log/wazuh-install.log  
07/08/2026 23:03:57 INFO: --- Dependencies ---  
07/08/2026 23:03:57 INFO: Installing gawk.  
07/08/2026 23:04:03 INFO: Verifying that your system meets the recommended minimum hardware requirements.  
07/08/2026 23:04:03 INFO: Wazuh web interface port will be 443.  
07/08/2026 23:04:12 INFO: --- Dependencies ---  
07/08/2026 23:04:12 INFO: Installing debhelper.  
07/08/2026 23:05:16 INFO: Wazuh repository added.  
07/08/2026 23:05:16 INFO: --- Configuration files ---  
07/08/2026 23:05:16 INFO: Generating configuration files.  
07/08/2026 23:05:17 INFO: Generating the root certificate.  
07/08/2026 23:05:17 INFO: Generating Admin certificates.  
07/08/2026 23:05:17 INFO: Generating Wazuh Indexer certificates.  
07/08/2026 23:05:17 INFO: Generating Filebeat certificates.  
07/08/2026 23:05:18 INFO: Generating Wazuh dashboard certificates.  
07/08/2026 23:05:18 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary  
for installation.  
07/08/2026 23:05:18 INFO: --- Wazuh Indexer ---  
07/08/2026 23:05:19 INFO: Starting Wazuh Indexer installation.  
07/08/2026 23:10:14 INFO: Wazuh Indexer installation finished.  
07/08/2026 23:10:15 INFO: Wazuh Indexer post-install configuration finished.  
07/08/2026 23:10:15 INFO: Starting service wazuh-indexer.  
07/08/2026 23:10:46 INFO: Wazuh-indexer service started.  
07/08/2026 23:10:46 INFO: Initializing Wazuh Indexer cluster security settings.  
07/08/2026 23:10:51 INFO: Wazuh Indexer cluster security configuration initialized.  
07/08/2026 23:10:51 INFO: Wazuh Indexer cluster initialized.  
07/08/2026 23:10:51 INFO: --- Wazuh server ---  
07/08/2026 23:10:51 INFO: Starting the Wazuh manager installation.  
07/08/2026 23:14:26 INFO: Wazuh manager installation finished.  
07/08/2026 23:14:26 INFO: Wazuh manager vulnerability detection configuration finished.  
07/08/2026 23:14:26 INFO: Starting service wazuh-manager.  
07/08/2026 23:14:44 INFO: Wazuh-manager service started.  
07/08/2026 23:14:44 INFO: Starting Filebeat installation.  
07/08/2026 23:15:07 INFO: Filebeat installation finished.
```

```
Activities Terminal Aug 7 23:21
sarra@sarra-VirtualBox: ~

for installation.
07/08/2026 23:05:18 INFO: --- Wazuh indexer ---
07/08/2026 23:05:19 INFO: Starting Wazuh Indexer installation.
07/08/2026 23:10:15 INFO: Wazuh Indexer installation finished.
07/08/2026 23:10:15 INFO: Wazuh Indexer post-install configuration finished.
07/08/2026 23:10:15 INFO: Starting service wazuh-indexer.
07/08/2026 23:10:46 INFO: Wazuh-Indexer service started.
07/08/2026 23:10:46 INFO: Initializing Wazuh Indexer cluster security settings.
07/08/2026 23:10:51 INFO: Wazuh Indexer cluster security configuration initialized.
07/08/2026 23:10:51 INFO: Wazuh Indexer cluster initialized.
07/08/2026 23:10:51 INFO: --- Wazuh server ---
07/08/2026 23:10:51 INFO: Starting the Wazuh manager installation.
07/08/2026 23:14:26 INFO: Wazuh manager installation finished.
07/08/2026 23:14:26 INFO: Wazuh manager vulnerability detection configuration finished.
07/08/2026 23:14:26 INFO: Starting service wazuh-manager.
07/08/2026 23:14:44 INFO: Wazuh-manager service started.
07/08/2026 23:14:44 INFO: Starting Filebeat installation.
07/08/2026 23:15:07 INFO: Filebeat installation finished.
07/08/2026 23:15:08 INFO: Filebeat post-install configuration finished.
07/08/2026 23:15:08 INFO: Starting service filebeat.
07/08/2026 23:15:10 INFO: filebeat service started.
07/08/2026 23:15:10 INFO: --- Wazuh dashboard ---
07/08/2026 23:15:10 INFO: Starting Wazuh dashboard installation.
07/08/2026 23:18:24 INFO: Wazuh dashboard installation finished.
07/08/2026 23:18:24 INFO: Wazuh dashboard post-install configuration finished.
07/08/2026 23:18:24 INFO: Starting service wazuh-dashboard.
07/08/2026 23:18:25 INFO: Wazuh-dashboard service started.
07/08/2026 23:18:28 INFO: Updating the internal users.
07/08/2026 23:18:35 INFO: A backup of the internal users has been saved in the /etc/wazuh-indexer/internalusers-backup folder.
07/08/2026 23:18:49 INFO: The filebeat.yml file has been updated to use the Filebeat Keystore username and password.
07/08/2026 23:19:51 INFO: Initializing Wazuh dashboard web application.
07/08/2026 23:19:52 INFO: Wazuh dashboard web application initialized.
07/08/2026 23:19:52 INFO: --- Summary ---
07/08/2026 23:19:52 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: *2pk0RuR+suJufauEbw.KFG0P00GXU4
07/08/2026 23:19:52 INFO: --- Dependencies ---
07/08/2026 23:19:52 INFO: Removing gawk.
07/08/2026 23:19:57 INFO: Installation finished.
sarra@sarra-VirtualBox: $
```

```
Activities Terminal Aug 8 13:14
sarra@sarra-VirtualBox: ~

07/08/2026 23:18:49 INFO: The filebeat.yml file has been updated to use the Filebeat Keystore username and password.
07/08/2026 23:19:51 INFO: Initializing Wazuh dashboard web application.
07/08/2026 23:19:52 INFO: Wazuh dashboard web application initialized.
07/08/2026 23:19:52 INFO: --- Summary ---
07/08/2026 23:19:52 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: *2pk0RuR+suJufauEbw.KFG0P00GXU4
07/08/2026 23:19:52 INFO: --- Dependencies ---
07/08/2026 23:19:52 INFO: Removing gawk.
07/08/2026 23:19:57 INFO: Installation finished.
sarra@sarra-VirtualBox: $ sudo systemctl status wazuh-manager
[sudo] password for sarra:
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-08-07 23:19:35 IST; 9min ago
     Tasks: 213 (limit: 7018)
    Memory: 828.1M
       CPU: 1min 8.260s
      CGroup: /system.slice/wazuh-manager.service
              └─98720 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/
                └─98721 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/
                  └─98722 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/
                    └─98725 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/
                      └─98728 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/
                        └─98764 /var/ossec/bin/wazuh-authd
                          └─98778 /var/ossec/bin/wazuh-db
                            └─98814 /var/ossec/bin/wazuh-execd
                              └─98828 /var/ossec/bin/wazuh-analysisd
                                └─98875 /var/ossec/bin/wazuh-syscheckd
                                  └─98892 /var/ossec/bin/wazuh-remoted
                                    └─98927 /var/ossec/bin/wazuh-logcollector
                                      └─98947 /var/ossec/bin/wazuh-monitord
                                        └─98969 /var/ossec/bin/wazuh-modulesd

Aug 07 23:19:29 sarra-VirtualBox env[98658]: Started wazuh-syscheckd...
Aug 07 23:19:30 sarra-VirtualBox env[98658]: Started wazuh-remoted...
Aug 07 23:19:31 sarra-VirtualBox env[98658]: Started wazuh-logcollector...
Aug 07 23:19:32 sarra-VirtualBox env[98658]: Started wazuh-monitord...
Aug 07 23:19:32 sarra-VirtualBox env[98966]: 2026/08/07 23:19:32 wazuh-modulesd...
Aug 07 23:19:32 sarra-VirtualBox env[98966]: 2026/08/07 23:19:32 wazuh-modulesd...
```

```
Activities Terminal Aug 8 13:15 sarra@sarra-VirtualBox: ~

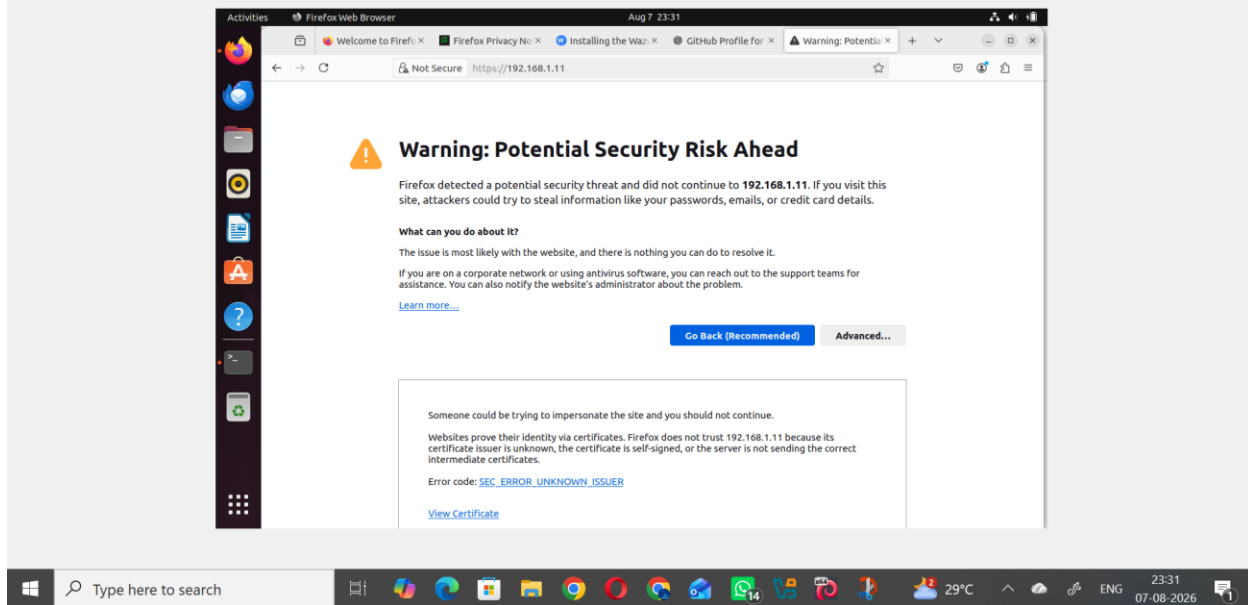
sarra@sarra-VirtualBox:~$ sudo systemctl status wazuh-indexer
● wazuh-indexer.service - wazuh-indexer
   Loaded: loaded (/lib/systemd/system/wazuh-indexer.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-08-07 23:10:46 IST; 19min ago
     Docs: https://documentation.wazuh.com
   Main PID: 45167 (java)
    Tasks: 72 (limit: 7018)
   Memory: 1.3G
      CPU: 1min 45.839s
   CGroup: /system.slice/wazuh-indexer.service
           └─45167 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -DopensearchDashboards

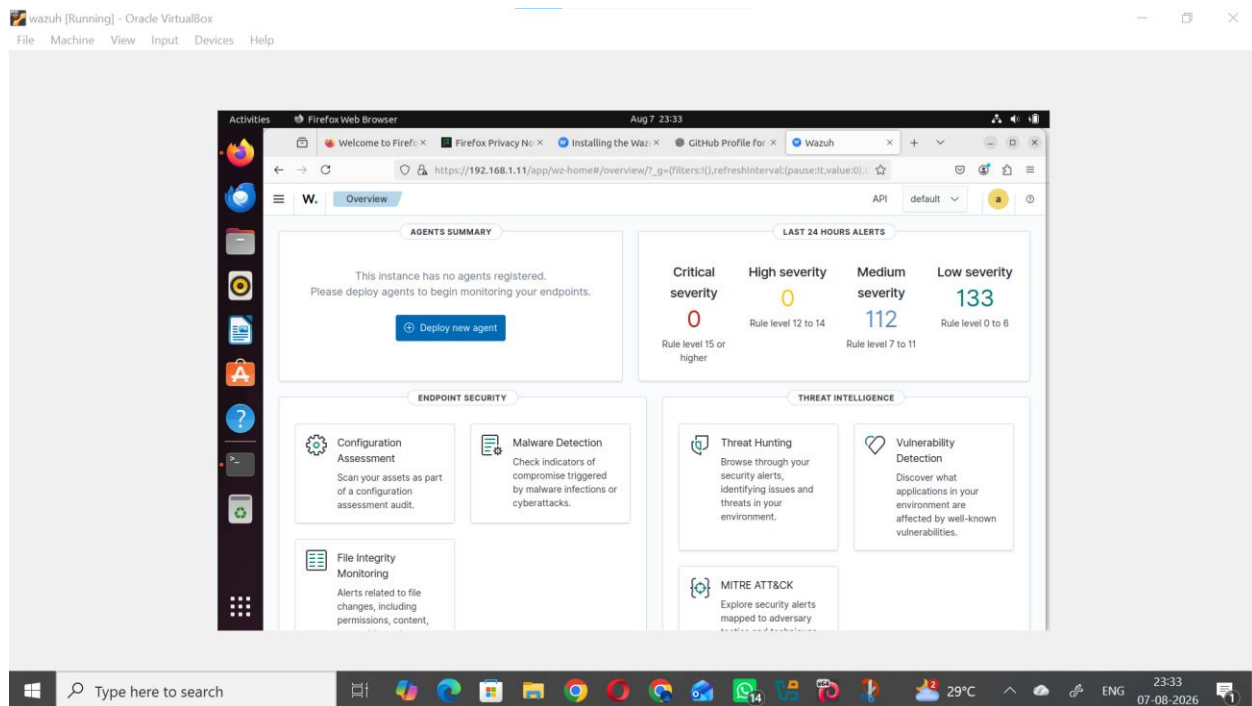
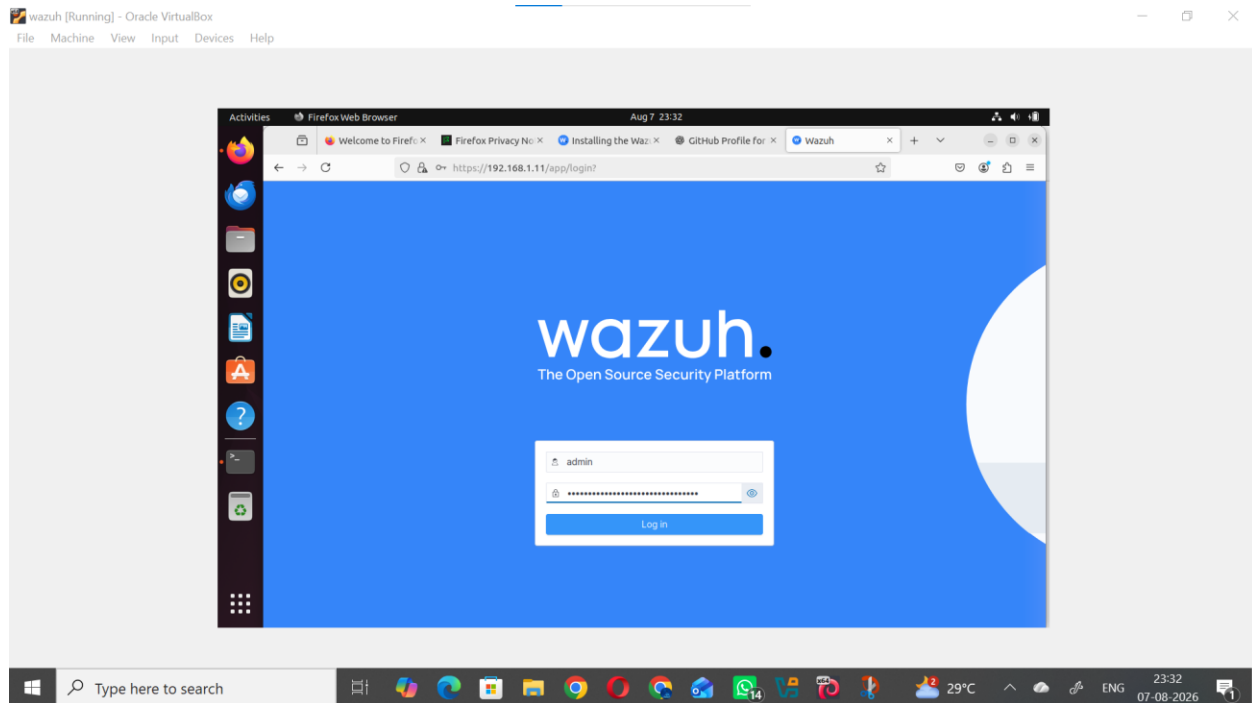
Aug 07 23:10:21 sarra-VirtualBox systemd-entrypoint[45167]: WARNING: System::se
Aug 07 23:10:21 sarra-VirtualBox systemd-entrypoint[45167]: WARNING: Please con
Aug 07 23:10:21 sarra-VirtualBox systemd-entrypoint[45167]: WARNING: System::se
Aug 07 23:10:24 sarra-VirtualBox systemd-entrypoint[45167]: Aug 07, 2026 11:10:p
Aug 07 23:10:24 sarra-VirtualBox systemd-entrypoint[45167]: WARNING: COMPAT Loca
Aug 07 23:10:24 sarra-VirtualBox systemd-entrypoint[45167]: WARNING: A terminal
Aug 07 23:10:24 sarra-VirtualBox systemd-entrypoint[45167]: WARNING: System::se
Aug 07 23:10:24 sarra-VirtualBox systemd-entrypoint[45167]: WARNING: Please con
Aug 07 23:10:24 sarra-VirtualBox systemd-entrypoint[45167]: WARNING: System::se
Aug 07 23:10:46 sarra-VirtualBox systemd[1]: Started wazuh-indexer.

sarra@sarra-VirtualBox:~$ sudo systemctl status wazuh-dashboard
● wazuh-dashboard.service - wazuh-dashboard
   Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-08-07 23:19:38 IST; 10min ago
     Main PID: 100253 (node)
    Tasks: 11 (limit: 7018)
   Memory: 156.4M
      CPU: 11.486s
   CGroup: /system.slice/wazuh-dashboard.service
           └─100253 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --

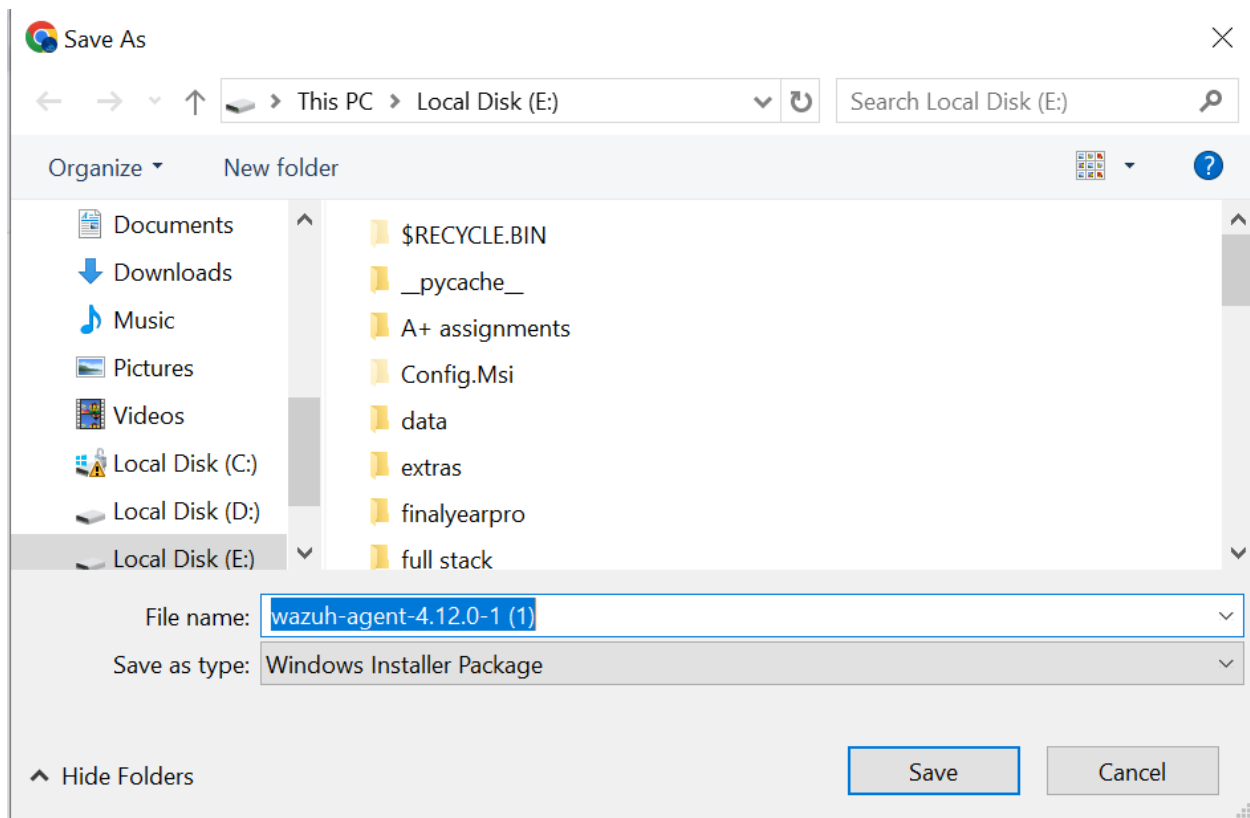
Aug 07 23:19:48 sarra-VirtualBox opensearch-dashboards[100253]: {"type":"log",
Aug 07 23:19:48 sarra-VirtualBox opensearch-dashboards[100253]: {"type":"log",
Aug 07 23:19:48 sarra-VirtualBox opensearch-dashboards[100253]: {"type":"log",
Aug 07 23:19:48 sarra-VirtualBox opensearch-dashboards[100253]: {"type":"log",
Aug 07 23:19:49 sarra-VirtualBox opensearch-dashboards[100253]: {"type":"log",
```

wazuh [Running] - Oracle VirtualBox
File Machine View Input Devices Help





Wazuh Agent on Windows



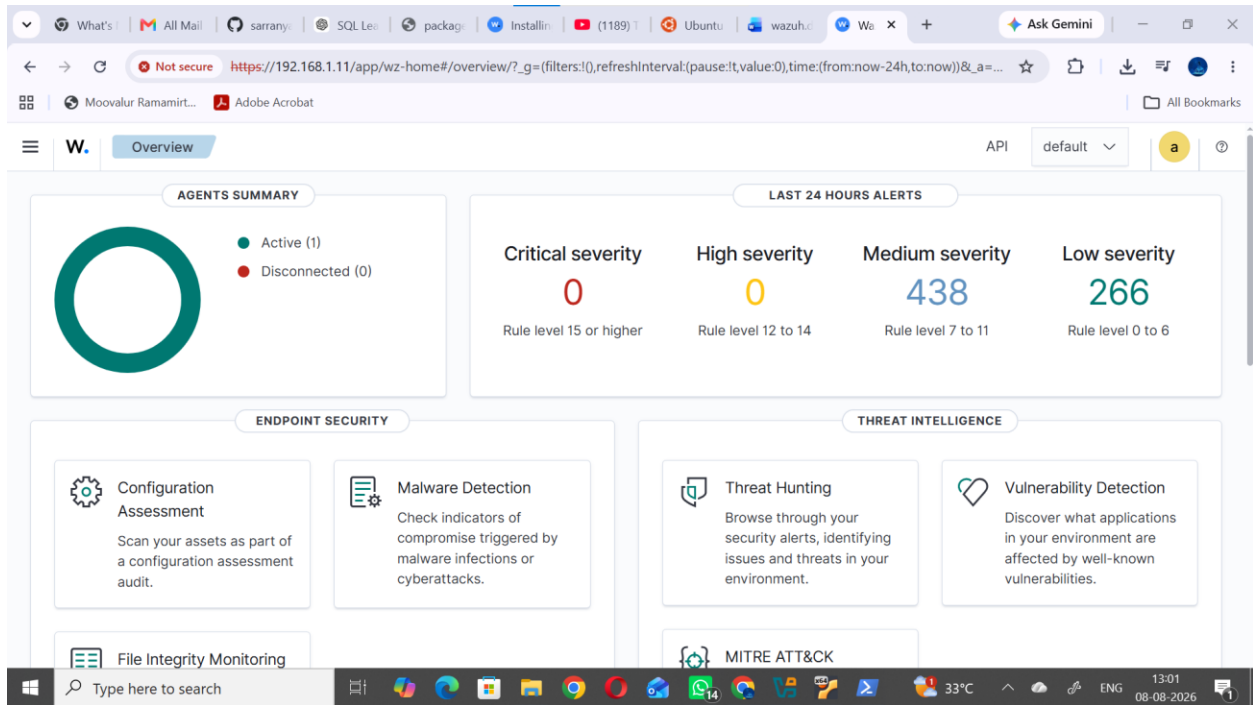
```
PS E:\> Test-Path "C:\Program Files (x86)\ossec-agent"
False
PS E:\> msixexec.exe /i "E:\wazuh-agent-4.14.7-1.msi" WAZUH_MANAGER="192.168.1.11"
PS E:\> Get-Service | Where-Object {$_.Name -like "*wazuh*" -or $_.DisplayName -like "*wazuh*"}

Status  Name      DisplayName
-----
Stopped WazuhSvc  Wazuh

PS E:\> Start-Service WazuhSvc
PS E:\> Get-Service WazuhSvc

Status  Name      DisplayName
-----
Running WazuhSvc  Wazuh

PS E:\>
```

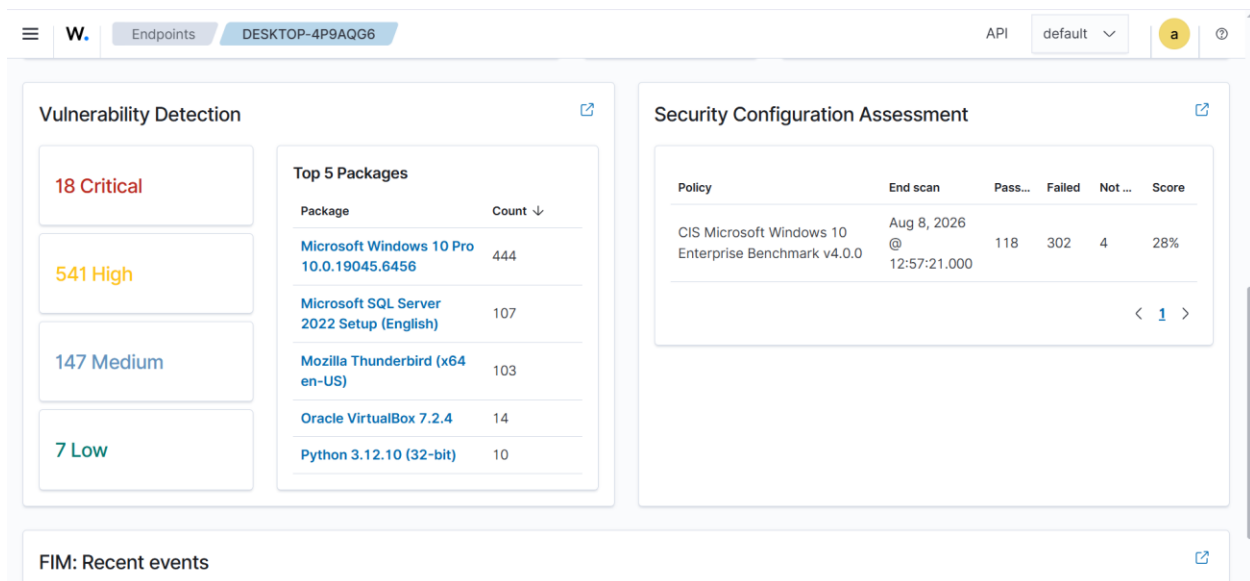
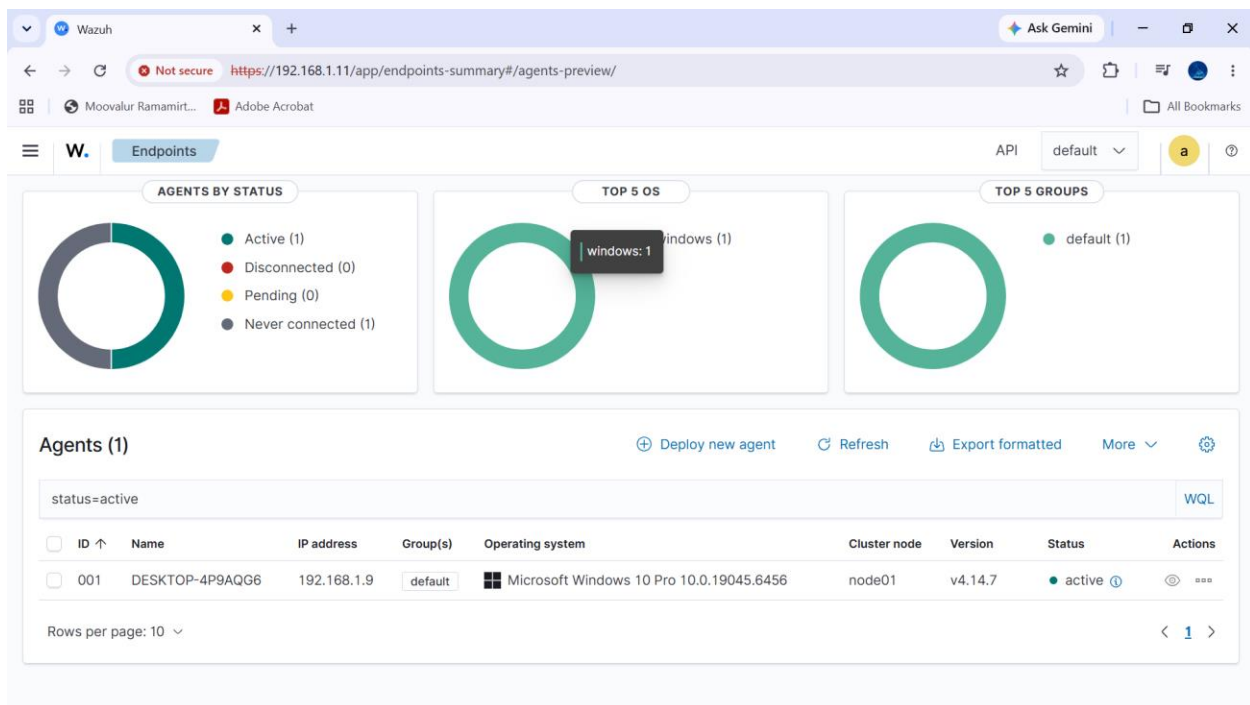



```
Activities Terminal Aug 8 13:29
sarra@sarra-VirtualBox: ~
*****
* Wazuh v4.14.7 Agent manager.
* The following options are available: *
*****
(A)dd an agent (A).
(E)xtract key for an agent (E).
(L)ist already added agents (L).
(R)emove an agent (R).
(Q)uit.
Choose your action: A,E,L,R or Q: A
- Adding a new agent (use '\q' to return to the main menu).
Please provide the following:
* A name for the new agent: WINDOWS_SGENT
* The IP Address of the new agent: 192.168.1.9
Confirm adding it?(y/n): y
Agent added with ID 002.

*****
* Wazuh v4.14.7 Agent manager.
* The following options are available: *
*****
(A)dd an agent (A).
(E)xtract key for an agent (E).
(L)ist already added agents (L).
(R)emove an agent (R).
(Q)uit.
Choose your action: A,E,L,R or Q: E
Available agents:
ID: 001, Name: DESKTOP-4P9AQG6, IP: any
ID: 002, Name: WINDOWS_SGENT, IP: 192.168.1.9
Provide the ID of the agent to extract the key (or '\q' to quit): 001

Agent key information for '001' is:
MDAxIERFU0tUT1AtNFASQVFHhbnkgMGUSYTNlNTE3ODhmNmFkMzE2OTFlOTliYmFLZjZkMWEyZDQzZGIxNmRjOGZlNmMwOWY0YjMSN2ESYzNjZGM3Yg==

** Press ENTER to return to the main menu.
```



File integrity Monitoring


```
PS E:\> notepad "C:\Program Files (x86)\ossec-agent\ossec.conf"
PS E:\>
```

```
<!-- File integrity monitoring -->
<syscheck>

  <disabled>no</disabled>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>

  <!-- Default files to be monitored. -->
  <directories recursion_level="0" restrict="regedit.exe|system.ini|win.ini">%WINDIR%</directories>

  <directories recursion_level="0" restrict="at.exe|attrib.exe|cacls.exe|cmd.exe|eventcreate.exe|ftp.exe|lsass.exe|net.exe|net1.exe|netsh.exe|reg
  <directories recursion_level="0">%WINDIR%\SysNative\drivers\etc</directories>
  <directories recursion_level="0" restrict="WMIC.exe">%WINDIR%\SysNative\wbem</directories>
  <directories recursion_level="0" restrict="powershell.exe">%WINDIR%\SysNative\WindowsPowerShell\v1.0</directories>
  <directories recursion_level="0" restrict="winrm.vbs">%WINDIR%\SysNative</directories>

  <!-- 32-bit programs. -->
  <directories recursion_level="0" restrict="at.exe|attrib.exe|cacls.exe|cmd.exe|eventcreate.exe|ftp.exe|lsass.exe|net.exe|net1.exe|netsh.exe|reg
  <directories recursion_level="0">%WINDIR%\System32\drivers\etc</directories>
  <directories recursion_level="0" restrict="WMIC.exe">%WINDIR%\System32\wbem</directories>
  <directories recursion_level="0" restrict="powershell.exe">%WINDIR%\System32\WindowsPowerShell\v1.0</directories>
  <directories recursion_level="0" restrict="winrm.vbs">%WINDIR%\System32</directories>

  <directories realtime="yes">%PROGRAMDATA%\Microsoft\Windows\Start Menu\Programs\Startup</directories>
  <directories realtime="yes">C:\Users\SKY\Downloads\WAZUH TEST</directories>
  <ignore>%PROGRAMDATA%\Microsoft\Windows\Start Menu\Programs\Startup\desktop.ini</ignore>
```

